Futuristic 3D Smart Automobile Service Garage | ROLL: 2107087

1. Scene Description

A futuristic automobile inspection garage is designed as a controlled maintenance environment where a vehicle enters a smart inspection chamber for automated servicing. The scene begins with a vehicle positioned outside the garage entrance while the inspection area remains illuminated by overhead lighting.

When the garage door opens, the vehicle enters and moves toward the hydraulic lift platform. After reaching the correct position, the door closes and the hydraulic lift raises the vehicle. A robotic inspection arm equipped with a moving spotlight performs an automated scanning process around the vehicle.

The scene demonstrates different computer graphics concepts including hierarchical modeling, geometric object generation, transformation, dynamic lighting, Gouraud shading, and Phong shading.

2. Scene Objects & Components

Component	Description	Animation
Automobile	Main object created using basic geometric primitives including body, wheels, windows, and headlights	Vehicle moves into the garage and wheels rotate during movement
Garage Entrance Door	Sliding door placed at the entrance of the inspection chamber	Opens before vehicle arrival and closes after vehicle enters
Hydraulic Lift Platform	Platform used to raise the vehicle during inspection	Moves upward and downward using transformation operations
Robotic Inspection Arm	Mechanical arm created using multiple primitive components with a scanner attached at the end	Arm joints rotate and scanner moves around the vehicle
Inspection Light Fixture	Spotlight attached to the robotic arm and ceiling lighting system reused as multiple instances	Spotlight follows arm movement and illuminates different areas
Garage Environment	Complete inspection chamber containing floor, walls, and ceiling created using basic geometric surfaces	Static environment providing the overall scene structure

3. Lighting Design (Light Sources in the Scene)

Light	Behavior	Purpose
Directional (ceiling)	Fixed direction	Provides overall illumination of the inspection chamber
Point Light (Vehicle Headlights)	Positioned inside each headlight	Light-inside-object, falloff demonstration

Spotlight (Inspection scanner)	Mounted on robotic arm and moves during inspection	Dynamic/moving light source
Ambient light	Base illumination applied throughout the environment	Maintains visibility of the objects throughout the scene

4. Overall Animation Flow

Step 1: Vehicle Arrival

- ☐ Garage door opens.
- ☐ Vehicle enters the inspection chamber.
- ☐ Wheels rotate while moving.
- ☐ Vehicle stops on the hydraulic lift.

Step 2: Vehicle Positioning

- ☐ Garage door closes.
- ☐ Vehicle aligns with the inspection platform.

Step 3: Automated Inspection

- ☐ Hydraulic lift raises the vehicle.
- ☐ Robotic arm starts scanning the vehicle.
- ☐ Inspection spotlight moves with the robotic arm.

Step 4: Lighting Demonstration

- ☐ Vehicle headlights turn on.
- ☐ Internal point lights illuminate nearby surfaces.
- ☐ Different lighting effects are observed under Gouraud and Phong shading.

Step 5: Service Completion

- ☐ Hydraulic lift lowers the vehicle.
- ☐ Garage door opens.
- ☐ Vehicle exits the inspection chamber.

5. Interactive Features

- ☐ Move the camera around the inspection garage.
- ☐ Turn vehicle headlights ON/OFF.
- ☐ Control the inspection spotlight movement.
- ☐ Rotate the inspection light source manually.
- ☐ Switch between Gouraud shading and Phong shading modes.
- ☐ Observe lighting differences on different materials.